WORKED SOLUTION

STEP III 2006, Question 12

Probability and Statistics • Normal Approximation Expectation

STEP III 2006, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s3-q12

The key idea

The extra bus is only worth what it earns on average, so the licence fee the company should consider is at most the *expected* extra profit per year. The extra bus earns nothing on a day when the existing fleet already copes, and it can never carry more than its 32 seats. So we need the expectation of a "clipped" random variable. The numbers are chosen so that, under the normal approximation, the old and new capacities land on very tidy points of the distribution.

Solution

Setting up the model

On a single arrival, let X be the number of the 1024 tourists directed to the bus station. Each tourist goes there with probability $\frac{1}{2}$, and we assume the tourists are directed independently of one another. Then

$$X \sim \mathrm{B}(1024, \tfrac{1}{2}), \qquad \mathrm{E}(X) = 512, \qquad \mathrm{Var}(X) = 1024 \cdot \tfrac{1}{2} \cdot \tfrac{1}{2} = 256.$$

Since n is large and $p = \frac{1}{2}$, we use the normal approximation $X \approx \mathrm{N}(512, 16^2)$, and write $X = 512 + 16Z$ with $Z \sim \mathrm{N}(0,1)$. As instructed, we make no continuity correction.

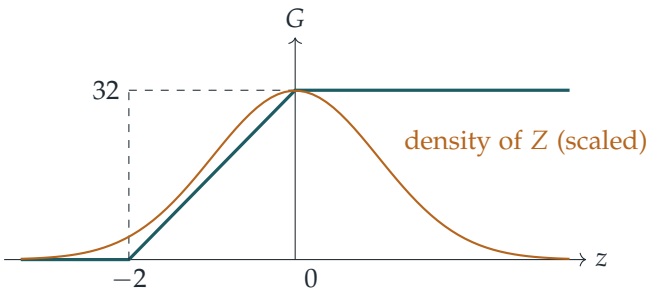
The 15 buses carry $15 \times 32 = 480$ passengers, so the current profit on one arrival is $£\min(X, 480)$. With a 16th bus the capacity is 512 and the profit is $£\min(X, 512)$. The extra profit on one arrival is therefore

$$G = \min(X,512) - \min(X,480) = \begin{cases} 0, & X \le 480, \\ X - 480, & 480 \le X \le 512, \\ 32, & X \ge 512. \end{cases}$$

In terms of Z, the two break points are $X = 480 \iff Z = -2$ and $X = 512 \iff Z =$

0, and in the middle range $X - 480 = 32 + 16Z$. So

$$G = \begin{cases} 0, & Z \le -2, \\ 32 + 16Z, & -2 \le Z \le 0, \\ 32, & Z \ge 0. \end{cases}$$



The expected extra profit

Let $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$ be the standard normal density. Splitting the expectation over the three ranges,

$$\mathrm{E}(G) = 32\,\mathrm{P}(Z \ge 0) + \int_{-2}^{0} (32 + 16z)\,\phi(z)\,\mathrm{d}z.$$

The first term is $32 \times \frac{1}{2} = 16$. For the integral, the constant part gives

$$32 \int_{-2}^{0} \phi(z)\,\mathrm{d}z = 32(\Phi(0) - \Phi(-2)) = 32(\tfrac{1}{2} - (1 - \Phi(2))) = 32\Phi(2) - 16,$$

using the symmetry $\Phi(-2) = 1 - \Phi(2)$.

Ask yourself How do we integrate $z\,\phi(z)$ without a table?

Differentiate the density: $\phi'(z) = -z\,\phi(z)$. So $z\,\phi(z)$ has the exact antiderivative $-\phi(z)$, and

$$16 \int_{-2}^{0} z\,\phi(z)\,\mathrm{d}z = 16[-\phi(z)]_{-2}^{0} = -16(\phi(0) - \phi(2)) = -\frac{16}{\sqrt{2\pi}}(1 - e^{-2}).$$

(It is negative, as it should be: $z < 0$ throughout the range.)

Adding the three pieces, the 16s cancel:

$$\mathrm{E}(G) = 16 + (32\Phi(2) - 16) - \frac{16}{\sqrt{2\pi}}(1 - e^{-2}) = 32\Phi(2) - \frac{16}{\sqrt{2\pi}}(1 - e^{-2}).$$

The licence fee

There are 50 arrivals a year, so by linearity of expectation the expected extra annual profit from the 16th bus is

$$50\,\mathrm{E}(G) = 1600\,\Phi(2) - \frac{800}{\sqrt{2\pi}}(1 - \mathrm{e}^{-2}) \text{ pounds.}$$

Paying a licence fee larger than this would, on average, lose the company money, while any smaller fee leaves an expected gain. So this is (approximately, because of the normal approximation) the largest annual fee worth considering, which is the required result. Numerically it is about £1288.

Checks

- Sanity bounds: the extra bus earns between 0 and 32 per trip, so the annual figure must lie between 0 and 1600. Our value $1600 \times 0.97725 - 319.15 \times 0.86466 \approx 1287.6$ does.
- Exact computation: summing $\min(k, 512) - \min(k, 480)$ against the exact $\mathrm{B}(1024, \frac{1}{2})$ probabilities and multiplying by 50 gives 1287.69, agreeing with the normal approximation 1287.64 to within 6p.

Common slips

- Taking the extra profit as $32\,\mathrm{P}(X > 480)$, as if the new bus always fills whenever it is needed. On days with between 480 and 512 bus passengers it is only partly full.
- Sign errors in $\int_{-2}^{0} z\phi(z)\,\mathrm{d}z$: the answer must be negative, since the integrand is.

Beyond STEP

The quantity $\mathrm{E}\big(\min(X, c)\big)$ is the expected sales of a newsvendor with stock c, and the "newsvendor problem" of choosing c is a classic of operations research; the same clipped payoff $\min(\max(X - a, 0), b - a)$ is a "call spread" in financial mathematics. The trick $\phi'(z) = -z\phi(z)$ is the start of the theory of truncated normal moments. Grimmett and Welsh, *Probability: An Introduction*, covers the normal approximation to the binomial and expectations of functions of random variables.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.